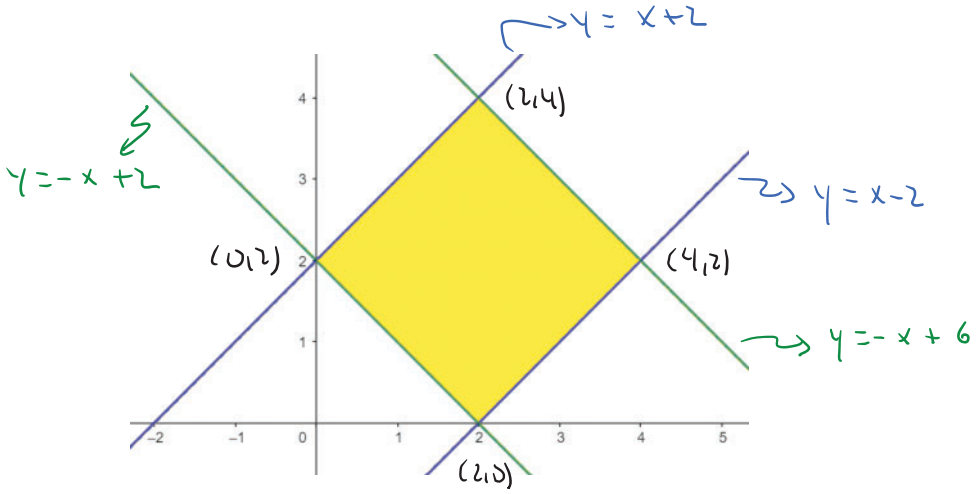
Problema 1 (primer parcial)

domingo, 22 de diciembre de 2024 19:58

Calcula la integral $\iint_R \frac{x-y}{\sqrt{13+x^2-y^2}}dxdy$, donde $R$ es el cuadrilátero de vértices $(2,0)$, $(4,2)$, $(2,4)$ y $(0,2)$ y proporciona un resultado numérico con un mínimo de dos decimales.




Tenemos las siguientes rectas: $x+y=2$, $x+y=6$, $x-y=2$, $x-y=-2$

Planteamos el siguiente cambio de variables:

$$\left.\begin{array}{l} u = x+y \\ v = x-y \end{array}\right] \quad J = \frac{1}{\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}} = \frac{1}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}} = \frac{1}{-2} \Rightarrow |J| = \frac{1}{2}$$

$$\iint_R \frac{x-y}{\sqrt{13+(x+y)(x-y)}}\,dx\,dy = \int_{u=2}^{u=6}\left(\int_{v=-2}^{v=2}\frac{v}{\sqrt{13+u\cdot v}}\,\frac{1}{2}\,dv\right)du =$$

$$= \frac{1}{2}\int_{v=-2}^{v=2}\left(\int_{u=2}^{u=6} v\cdot(13+u\cdot v)^{-1/2}\,du\right)dv = \frac{1}{2}\int_{v=-2}^{v=2} v\cdot\left[\frac{(13+u\cdot v)^{1/2}}{1/2\cdot v}\right]_{u=2}^{u=6}dv =$$

$$= \int_{v=-2}^{v=2}\frac{v}{v}\cdot\left((13+6v)^{1/2}-(13+2v)^{1/2}\right)dv =$$

$$= \left[\frac{(13+6v)^{3/2}}{3/2\cdot 6}\right]_{-2}^{2} - \left[\frac{(13+2v)^{3/2}}{3/2\cdot 2}\right]_{-2}^{2} = \frac{1}{9}\left[(13+6v)^{3/2}\right]_{-2}^{2} - \frac{1}{3}\left[(13+2v)^{3/2}\right]_{-2}^{2} =$$

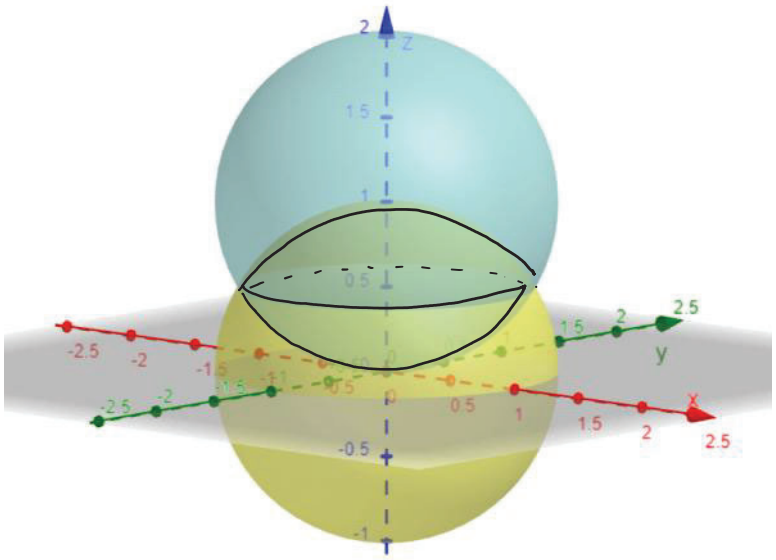
$$= \frac{1}{9}\left(\sqrt{125^3} - \sqrt{1^3}\right) - \frac{1}{3}\left(\sqrt{17^3} - \sqrt{9^3}\right) =$$

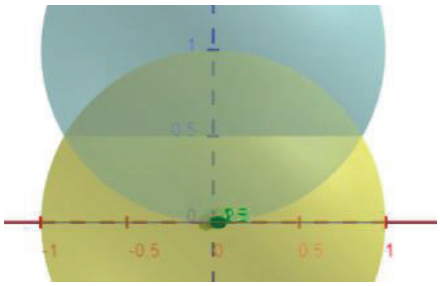
$$= \frac{1}{9}\left(125 - 1\right) - \frac{1}{3}\left(17\sqrt{17} - 27\right) = \boxed{\frac{205 - 51\sqrt{17}}{9} \simeq -0{,}59}$$

Problema 2 (primer parcial)

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Calcula el volumen del sólido interior a las esferas $x^2 + y^2 + z^2 = a^2$ y $x^2 + y^2 + (z - a)^2 = a^2$.



- Ecuación de la esfera de centro $(0,0,a)$ en coordenadas esféricas:

$$x^2 + y^2 + (z-a)^2 = a^2 \Rightarrow x^2 + y^2 + z^2 - 2az = 0 \Rightarrow \rho^2 = 2a\rho\cos(\varphi) \Rightarrow \rho = 2a\cos(\varphi)$$

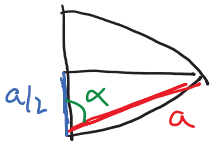
- Corte de las dos esferas:

$$x^2 + y^2 + z^2 - a^2 = x^2 + y^2 + (z-a)^2 - a^2 \Rightarrow z^2 = z^2 - 2az + a^2 \Rightarrow z = a/2$$

- Ecuación $z = a/2$ en coordenadas cilíndricas:

$$z = \frac{a}{2} \Rightarrow \rho \cdot \cos(\varphi) = \frac{a}{2} \Rightarrow \rho = \frac{a}{2\cdot\cos(\varphi)}$$

- Cálculo del ángulo $\alpha$:




$$\cos(\alpha) = \frac{a/2}{a} \Rightarrow \alpha = \arccos(1/2) = \frac{\pi}{3}$$

$$V = 2\cdot\int_{\theta=0}^{\theta=2\pi}\left(\int_{\varphi=0}^{\varphi=\frac{\pi}{3}}\left(\int_{\rho=\frac{a}{2\cos(\varphi)}}^{\rho=a}\rho^2\cdot\operatorname{sen}(\varphi)\cdot d\rho\right)d\varphi\right)d\theta =$$

$$= 2 \cdot \int_{\theta=0}^{\theta=2\pi} \left( \int_{\varphi=0}^{\varphi=\frac{\pi}{3}} \operatorname{sen}(\varphi) \cdot \left[ \frac{\rho^3}{3} \right]_{\rho=\frac{a}{2\cos(\varphi)}}^{\rho=a} d\varphi \right) d\theta =$$

$$= \frac{2}{3} \int_{\theta=0}^{\theta=2\pi} \left( \int_{\varphi=0}^{\varphi=\frac{\pi}{3}} \operatorname{sen}(\varphi) \cdot \left( a^3 - \frac{a^3}{8\cos^3(\varphi)} \right) d\varphi \right) d\theta =$$

$$= \frac{2}{3} a^3 \left( \int_{\theta=0}^{\theta=2\pi} d\theta \right) \cdot \left( \int_{\varphi=0}^{\varphi=\pi/3} \left( \operatorname{sen}(\varphi) - \frac{1}{8} \operatorname{sen}(\varphi) \cdot \cos^{-3}(\varphi) \right) d\varphi \right) =$$

$$= \frac{2}{3} \cdot a^3 \cdot [\theta]_0^{2\pi} \left[ -\cos(\varphi) + \frac{1}{8} \frac{\cos^{-2}(\varphi)}{-2} \right]_0^{\pi/3} = \frac{4\pi a^3}{3} \left[ -\cos(\varphi) - \frac{1}{16} \frac{1}{\cos^2(\varphi)} \right]_0^{\pi/3} =$$

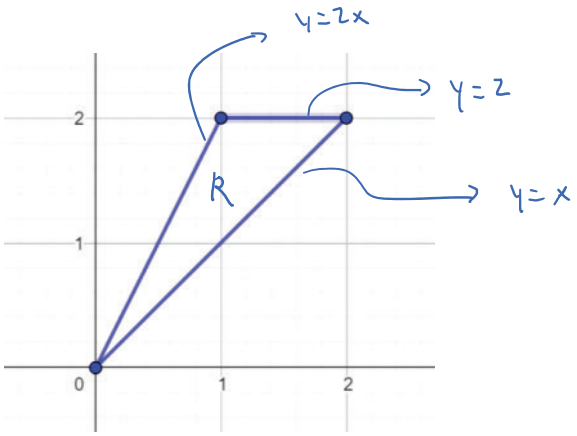
$$= \frac{4\pi a^3}{3} \left( \left( -\frac{1}{2} - \frac{1}{16} \frac{1}{(1/2)^2} \right) - \left( -1 - \frac{1}{16} \right) \right) = \frac{4\pi a^3}{3} \left( -\frac{1}{2} - \frac{1}{4} + 1 + \frac{1}{16} \right) =$$

$$= \frac{4\pi a^3}{3} \left( \frac{1}{2} - \frac{3}{16} \right) = \boxed{\frac{5\pi a^3}{12}}$$

Problema 3 (primer parcial)

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Supongamos que tenemos una lámina que se puede modelar como el recinto limitado por las rectas $y = 2$, $y = x$ e $y = 2x$. Si su densidad está definida por la función $\mu(x, y) = 4x + 2y + 6$, calcula su centro de masas. Proporciona el resultado como expresión exacta y adicionalmente en forma decimal aproximada.




- $m(R) = \iint_R \mu(x,y)\,dx\,dy = \iint_R (4x+2y+6)\,dx\,dy = \int_{y=0}^{y=2}\left(\int_{x=y/2}^{x=y}(4x+2y+6)\,dx\right)dy =$

$= \int_{y=0}^{y=2}\left[2x^2+(2y+6)x\right]_{x=y/2}^{x=y}dy = \int_{y=0}^{y=2}\left(\left(2y^2+2y^2+6y\right)-\left(\frac{y^2}{2}+y^2+3y\right)\right)dy =$

$= \int_{y=0}^{y=2}\left(\frac{5y^2}{2}+3y\right)dy = \left[\frac{5y^3}{6}+\frac{3y^2}{2}\right]_0^2 = \frac{20}{3}+6 = \frac{38}{3}$

- $M_x(R) = \iint_R y\cdot(4x+2y+6)\,dx\,dy = \int_{y=0}^{y=2}\left(\int_{x=y/2}^{x=y}\left(4yx+\left(2y^2+6y\right)\right)dx\right)dy =$

$= \int_{y=0}^{y=2}\left[2yx^2+\left(2y^2+6y\right)x\right]_{y/2}^{y}dy = \int_{y=0}^{y=2}\left(\left(2y^3+2y^3+6y^2\right)-\left(\frac{y^3}{2}+y^3+3y^2\right)\right)dy =$

$= \int_{y=0}^{y=2}\left(\frac{5}{2}y^3+3y^2\right)dy = \left[\frac{5}{2}\frac{y^4}{4}+y^3\right]_0^2 = \frac{5}{2}\cdot 4+8 = 18$

- $M_y(R) = \iint_R x(4x+2y+6)\,dx\,dy = \int_{y=0}^{y=2}\left(\int_{x=y/2}^{x=y}\left(4x^2+(2y+6)x\right)dx\right)dy =$

$= \int_{y=0}^{y=2}\left[\frac{4x^3}{3}+(2y+6)\frac{x^2}{2}\right]_{y/2}^{y}dy = \int_{y=0}^{y=2}\left(\left(\frac{4}{3}y^3+y^3+3y^2\right)-\left(\frac{y^3}{6}+\frac{y^3}{4}+\frac{3y^2}{4}\right)\right)dy =$

$$= \int_{y=0}^{y=2} \left( \frac{23}{12} y^3 + \frac{9}{4} y^2 \right) dy = \left[ \frac{23}{12} \frac{y^4}{4} + \frac{9}{4} \frac{y^3}{3} \right]_0^2 = \frac{92}{12} + 6 = \frac{164}{12} = \frac{41}{3}$$

Por lo tanto, $(\bar{x}, \bar{y}) = \left( \frac{M_y(R)}{m(R)}, \frac{M_x(R)}{m(R)} \right) = \left( \frac{41/3}{38/3}, \frac{18}{38/3} \right) = \boxed{\left( \frac{41}{38}, \frac{27}{19} \right) \simeq (1'08, 1'42)}$